Kindred AI: Agentic Accessibility & Learning

Empowering neurodivergent minds through Retrieval-Augmented Generation (RAG) and multi-agent AI systems designed for cognitive diversity.

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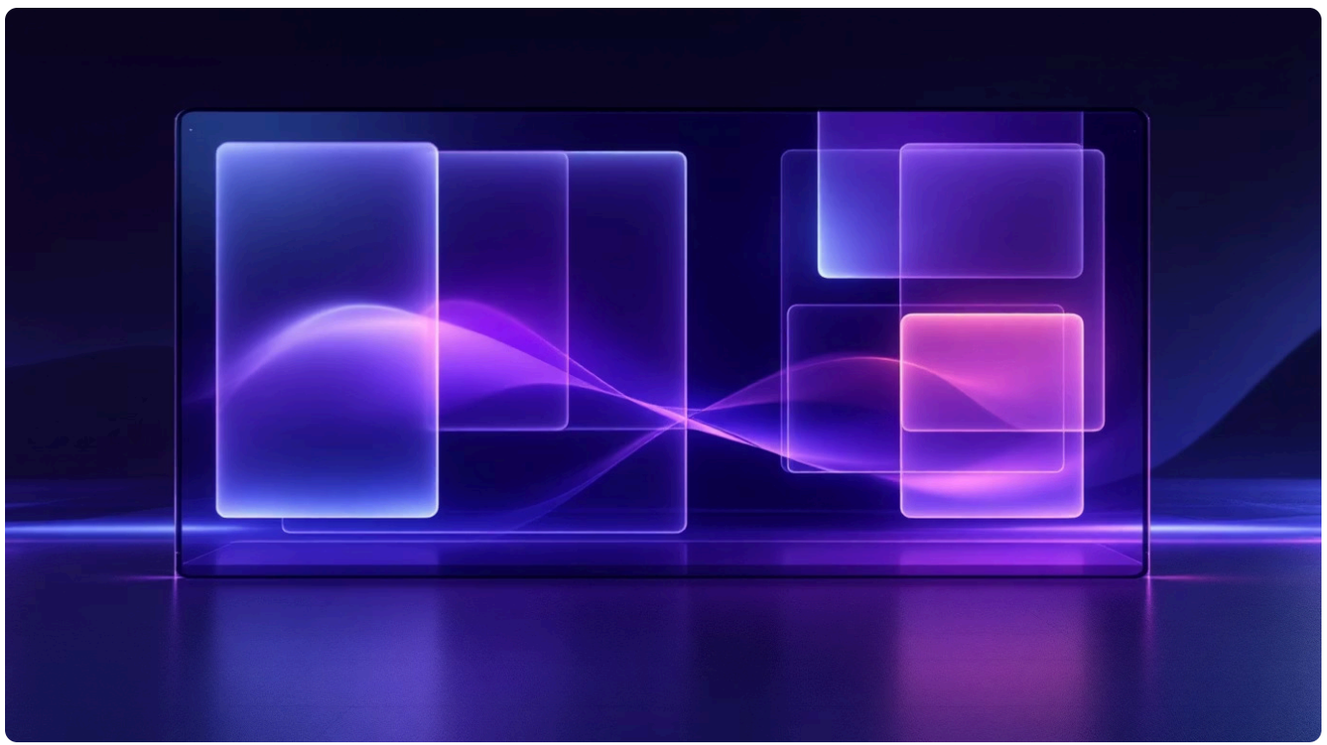
The Design Philosophy: Glassmorphism & Cognitive Load

Visual Design Principles

Kindred AI employs Glassmorphism interfaces with frosted glass panels, subtle depth, and ambient lighting to reduce cognitive load whilst maintaining aesthetic sophistication.

The interface balances Deep Space (#0F1115) backgrounds for low eye strain with Amethyst Purple (#A855F7) accents for sensory focus and navigation cues.



Dual-Panel Study Hub

Reader panel for content consumption alongside Sculptor panel for active learning and content manipulation

Cognitive Load Reduction

Strategic use of whitespace, consistent navigation patterns, and progressive disclosure of complex features

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The Accessibility Toolbar: Pill Dock

A customizable, always-accessible toolbar providing instant control over reading conditions and visual preferences.

OpenDyslexic Font

Built-in support for OpenDyslexic typeface with weighted bottoms to reduce letter confusion and improve readability for dyslexic users

Bionic Reading Logic

Automatically bolds the first few characters of each word to guide eye movement and accelerate comprehension through predictive reading patterns

Adjustable Line Spacing

Dynamic control over paragraph spacing, letter spacing, and word spacing to accommodate individual visual processing preferences

Dark Theme

Deep Space background with Slate-200 text for minimal eye strain during extended sessions

Crème Theme

Soft beige backgrounds with high contrast text for users sensitive to dark interfaces

Hi-Con Yellow/Black

Maximum contrast with yellow text on black background for optimal visual distinction

Focus Blue

Blue-dominant palette designed to enhance concentration and reduce distraction through color psychology

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Technical Architecture: The RAG Engine

Kindred AI leverages Retrieval-Augmented Generation (RAG) to combine semantic search with generative AI, ensuring responses are grounded in source material whilst maintaining contextual understanding.

Ingestion

Documents, PDFs, and web content are parsed and normalized into processable formats

Vectorization

ChromaDB converts text into high-dimensional vectors for semantic similarity matching

Semantic Retrieval

Context-aware search identifies relevant document chunks based on meaning, not keywords

Grounded Generation

Gemini API generates responses anchored to retrieved source material with citations

Optimization Detail: 2-sentence chunking with 15% overlap ensures context retention whilst enabling precise citation attribution

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The Agentic Layer: Multi-Agent System

Three specialized AI agents collaborate to transform raw content into accessible, ADHD-friendly learning material through coordinated workflow.

1

The Ingestion Specialist

Function: OCR processing, Markdown parsing, and document normalization

- Extracts text from scanned documents and images
- Parses structured formats (PDF, DOCX, EPUB)
- Normalizes encoding and removes artifacts

2

The Knowledge Sculptor

Function: Document Q&A, semantic search, and content organization

- Answers questions using source material
- Creates topic hierarchies and connections
- Generates summaries and key concepts

3

The Simplification Agent

Function: ADHD-friendly content rewriting and complexity reduction

- Breaks complex sentences into shorter units
- Adds concrete examples and analogies
- Highlights key information with visual markers

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Neuro-Inclusive Multimedia Experience

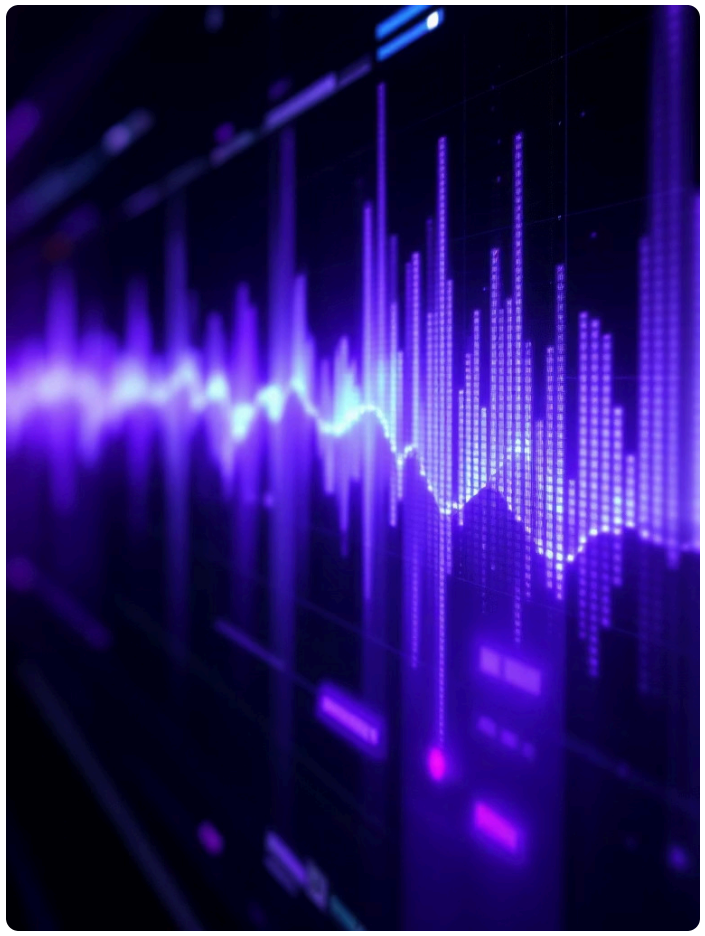
Voice Assistant Features

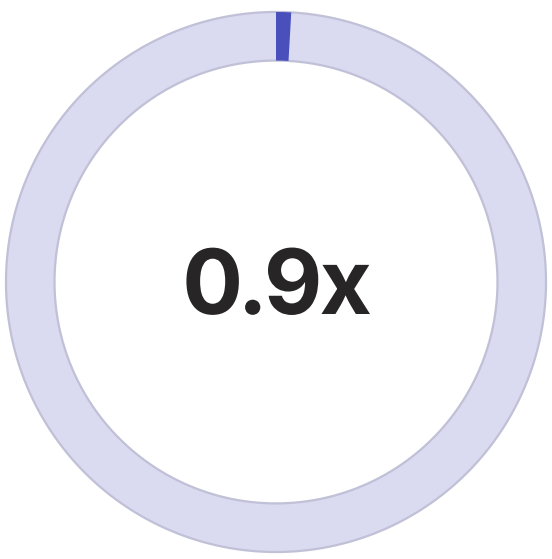
The integrated voice assistant operates at 0.9x playback speed by default, calibrated for improved auditory processing and comprehension.

32-bar visual waveforms provide sensory anchoring, synchronizing audio with text chunks to reinforce multimodal learning pathways.

Key Benefits

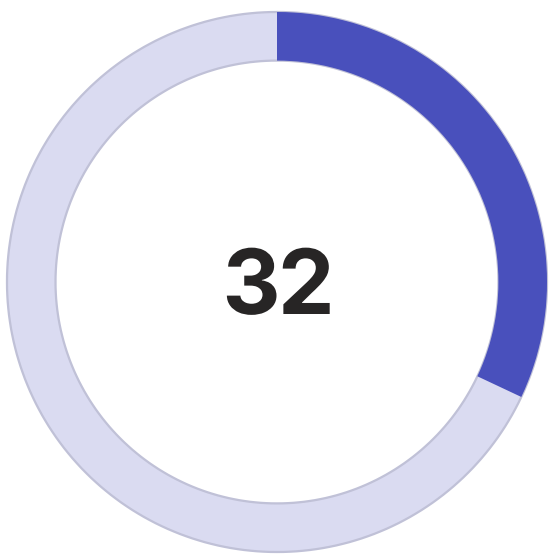
- Reduced processing load through slower, deliberate speech patterns
- Visual feedback maintains engagement during auditory input
- Text-audio synchronization supports working memory retention





Playback Speed

Optimized for neurodivergent processing



Visual Bars

Waveform provides sensory anchor

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Tech Stack & Export Capabilities

Frontend Stack

React with Next.js for server-side rendering, Framer Motion for animations, and Tailwind CSS for responsive design

Intelligence Layer

LangChain for agent orchestration, Gemini API for generative capabilities, and ChromaDB for vector storage

Export Formats

Kindred AI generates ADHD-friendly Markdown with simplified structure and Simplified PDFs optimized for readability and offline use.

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Empowering Neurodivergent Learning

Grounded AI responses

RAG architecture ensures all generated content is anchored to source material with proper citation

Customizable interface

Real-time adjustments to typography, spacing, and color schemes based on individual preferences

Multimodal engagement

Synchronized text, audio, and visual elements support diverse processing styles

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